

Energy Consumption Classification & Prediction

CSCI-6767 Applied Machine Learning — Course Project 2

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1. Motivation

Buildings account for approximately 40% of total energy consumption in developed countries. Accurately predicting whether a building will be a high or low energy consumer—and estimating the exact kilowatt-hours—is a critical problem for utility companies, building operators, and policymakers. Classification models can triage buildings into risk categories, while regression models provide continuous predictions for operational planning.

In this project, we apply four classification methods (Logistic Regression, Linear Discriminant Analysis, Quadratic Discriminant Analysis, and Gaussian Naïve Bayes) and two regression methods (Linear Regression and Poisson Regression) to a Kaggle energy consumption dataset. We evaluate each model using ROC curves, AUC, and standard classification metrics, and provide detailed statistical inference to understand which building features drive energy consumption.

2. Dataset

The dataset was sourced from Kaggle (Govinda Ram Sriram, “Energy Consumption Dataset”) and contains 1,100 building-day observations split into 1,000 training and 100 test samples. There are six predictor features and one continuous target variable.

Feature	Type	Range	Description
Building Type	Categorical	3 categories	Residential / Commercial / Industrial
Square Footage	Continuous	560 – 49,997	Building floor area in sq ft
Num. Occupants	Integer	1 – 99	Number of people in building
Appliances Used	Integer	1 – 49	Active appliances
Avg. Temperature	Continuous	10 – 35 °C	Ambient temperature
Day of Week	Binary	2 values	Weekday vs Weekend

Table 1. Feature descriptions for the energy consumption dataset.

The target variable, Energy Consumption, ranges from 1,684 to 6,531 kWh. For classification, we created a binary target using the training median (4,176 kWh) as the threshold and a three-class target using tertile boundaries (Low < 3,733, Medium: 3,733–4,638, High > 4,638 kWh).

3. Preprocessing

Label Encoding. Categorical features (Building Type and Day of Week) were converted to integers using scikit-learn’s LabelEncoder: Commercial → 0, Industrial → 1, Residential → 2; Weekday → 0, Weekend → 1. Encoders were fitted on training data only.

Standard Scaling. Continuous features were standardized to zero mean and unit variance: $z = (x - \mu) / \sigma$. Scalers were fitted exclusively on training data to prevent data leakage.

Target Engineering. For binary classification, observations above the training median (4,176 kWh) were labeled HIGH (1), below as LOW (0). For multi-class, tertile boundaries split data into three roughly equal groups.

4. Methods

4.1 Binary Logistic Regression

Logistic Regression models the probability of HIGH energy via the sigmoid function: $P(Y=1|X) = 1 / (1 + e^{-(\beta_0 + \beta_1 X_1 + \dots + \beta_6 X_6)})$. The model computes $z = \beta_0 + \beta_1 X_1 + \dots + \beta_6 X_6$, then passes it through the

sigmoid to get a probability. Coefficients are estimated via Maximum Likelihood Estimation (MLE). We used statsmodels for statistical inference and scikit-learn with L2 regularization for prediction.

Understanding the Statistical Output

Standard Error (SE) measures how precisely each coefficient is estimated — a low SE relative to the coefficient means a stable, reliable estimate. **The Z-statistic** is simply the coefficient divided by its SE ($Z = \text{coeff} / \text{SE}$), telling us how many standard errors away from zero the coefficient is; values above 1.96 indicate statistical significance at 95% confidence. The P-value then answers: "If this feature had zero effect, what is the probability of seeing a coefficient this large by chance?" For example, Square Footage has $Z = 14.76$ and $p \approx 0$, meaning it is an overwhelmingly significant predictor, while Average Temperature has $Z = -1.37$ and $p = 0.171$, meaning there is a 17.1% chance its effect is due to random noise — making it the only non-significant feature in our model.

Variable	Coeff	SE	Z-stat	P-value	Sig.	Odds Ratio
Sq Footage	3.731	0.253	14.76	2.59e-49	$p < 0.001$	41.7×
Occupants	1.532	0.153	10.00	1.47e-23	$p < 0.001$	4.6×
Appliances	1.260	0.141	8.95	3.68e-19	$p < 0.001$	3.5×
Bldg Type	-1.293	0.166	-7.80	6.33e-15	$p < 0.001$	0.27×
Day of Wk	-0.487	0.234	-2.08	0.038	$p < 0.05$	0.61×
Avg Temp	-0.161	0.118	-1.37	0.171	n.s.	0.85×

Table 2. Logistic Regression coefficients (statsmodels MLE)

4.2 Multi-class Logistic Regression

For three classes (Low/Medium/High), we trained multinomial logistic regression using the softmax function: $P(Y=k|X) = e^{(\beta_k X)} / \sum e^{(\beta_i X)}$. The model learns three weight vectors and normalizes them, so probabilities sum to 1. Evaluation uses One-vs-Rest (OvR) ROC analysis.

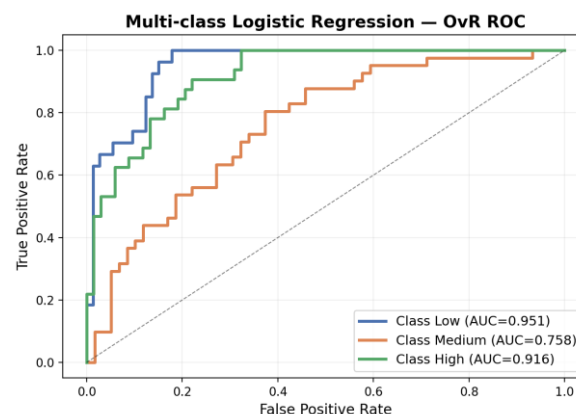


Figure 1. Multi-class Logistic Regression — One-vs-Rest ROC curves.

What the Per-Class AUC Values Mean

Low (AUC = 0.951): Excellent. The model easily identifies low-consumption buildings because they occupy one extreme of the energy distribution with distinctive features (small buildings, few occupants).

Medium (AUC = 0.758): Acceptable but notably lower. An AUC of 0.758 means the model correctly ranks a Medium building above a non-Medium building only 75.8% of the time. This class is hardest because it is defined by arbitrary tertile cutoffs in a continuous distribution, with no natural gap separating it from Low or High.

High (AUC = 0.916): Strong. Like Low, it occupies a distribution extreme with clear feature signatures (large buildings, many occupants and appliances).

4.3 Linear Discriminant Analysis (LDA)

LDA takes a generative approach: it models the feature distribution within each class as a multivariate Gaussian with shared covariance Σ , then uses Bayes' theorem to compute posteriors. The discriminant score $\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log(\pi_k)$ is linear in x , producing a straight-line decision boundary.

4.4 Quadratic Discriminant Analysis (QDA)

QDA relaxes the shared covariance assumption—each class gets its own Σ_k . This produces curved decision boundaries that can capture cases where classes have different shapes or spreads. The trade-off: more parameters to estimate, higher variance with small samples.

Defining the Effective Threshold

Rather than the default 0.5 threshold, we selected thresholds maximizing Youden's J statistic: $J = \text{TPR} - \text{FPR}$, the point on the ROC curve farthest from the diagonal. LDA's optimal threshold = 0.764; QDA's = 0.788. Both above 0.5, meaning the models require higher confidence before predicting HIGH—prioritizing precision over recall.

ROC Curve and AUC

The ROC curve plots True Positive Rate (sensitivity) against False Positive Rate (1 – specificity) at every threshold. A curve hugging the top-left corner indicates excellent discrimination; the diagonal represents random guessing.

AUC (Area Under the Curve) condenses the entire ROC curve into a single number. Its probabilistic interpretation: AUC is the probability that a randomly chosen HIGH building gets a higher predicted probability than a randomly chosen LOW building. AUC = 1.0 is perfect; 0.5 is random; > 0.9 is excellent.

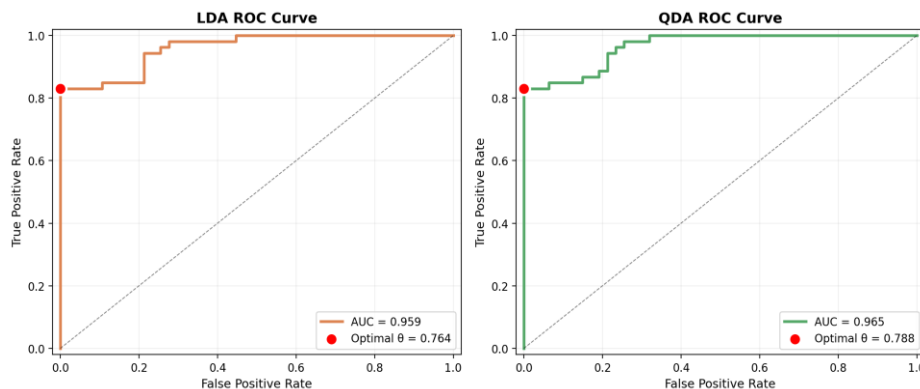


Figure 2. ROC curves for LDA (AUC = 0.960) and QDA (AUC = 0.965). Red dots mark optimal thresholds.

Outcome: Both curves hug the top-left corner, confirming excellent separation. QDA's slightly higher AUC (0.965 vs 0.960) suggests its flexible boundaries capture marginally more structure, but the small gap tells us the shared-covariance assumption of LDA is approximately valid—both classes spread out similarly in feature space.

4.5 Gaussian Naïve Bayes

Naïve Bayes assumes features are conditionally independent given the class, modeling each as a separate Gaussian per class. Despite this strong assumption, it performs competitively

when features are weakly correlated—as our $VIF \approx 1.0$ confirms. NB achieved $AUC = 0.954$ with a lower threshold (0.557), trading slightly lower precision (0.922) for higher recall (0.887).

4.6 Linear vs Poisson Regression

For continuous energy prediction, we compared Linear Regression (OLS) and Poisson Regression (log-link).

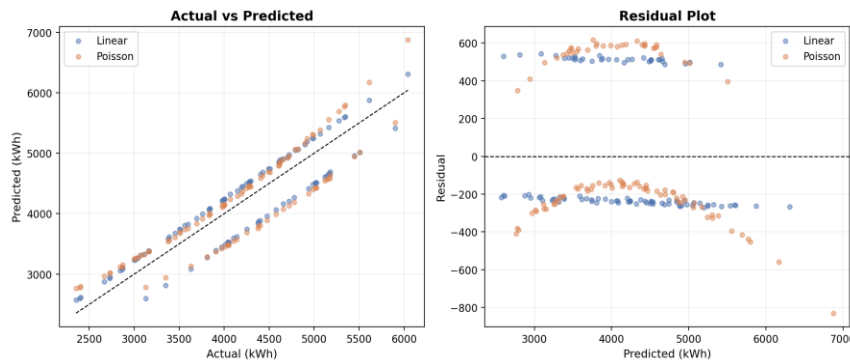


Figure 3. Actual vs Predicted scatter and residual analysis for Linear and Poisson Regression.

Outcome: Linear Regression outperforms ($R^2 = 0.814$ vs 0.778 , $RMSE = 357.1$ vs 390.4). Energy consumption is continuous, not count data, so OLS's identity link is the natural fit. Poisson's log-link introduces unnecessary nonlinearity.

5. Experiments & Results

5.1 Binary Classification Comparison

Model	Thresh.	Acc.	Prec.	Recall	F1	AUC	Rating
LR	0.779	0.91	1.000	0.830	0.907	0.963	Excellent
LDA	0.764	0.91	1.000	0.830	0.907	0.960	Excellent
QDA	0.788	0.91	1.000	0.830	0.907	0.965	Best
NB	0.557	0.90	0.922	0.887	0.904	0.954	Strong

Table 3. Binary classification results (100 test observations, optimal thresholds).

What These Results Indicate

AUC > 0.95 for all models: If we randomly pick one HIGH and one LOW building, each model correctly ranks the HIGH one above the LOW one more than 95% of the time. This confirms the six features carry strong signal for energy classification.

QDA achieves highest AUC (0.965): Its quadratic boundary captures slightly more structure than linear methods. However, the difference from LDA (0.005) is small, meaning a linear boundary is largely sufficient for this dataset.

Perfect precision (1.000) for LR, LDA, QDA: When these models predict HIGH, they are correct 100% of the time. No LOW building was misclassified as HIGH. This is achieved by high thresholds (0.76–0.79) that require strong confidence.

NB trades precision for recall: Its lower threshold (0.557) catches more actual HIGH buildings (recall = 0.887 vs 0.830) but introduces some false positives (precision = 0.922). This makes NB preferable if missing HIGH buildings is costlier than false alarms.

5.2 Regression Comparison

Metric	Linear Reg.	Poisson Reg.	Interpretation
R^2	0.814	0.778	81% variance explained
RMSE	357.1 kWh	390.4 kWh	Lower = better fit
MAE	332.1 kWh	347.6 kWh	Avg prediction error

Table 4. Regression performance on the test set.

$R^2 = 0.814$ means six features explain 81% of energy variance. The remaining 19% likely comes from factors not in the dataset (building age, insulation, HVAC type). R^2 above 0.8 is considered strong for real-world prediction.

6. Feature Importance & Confounding Analysis

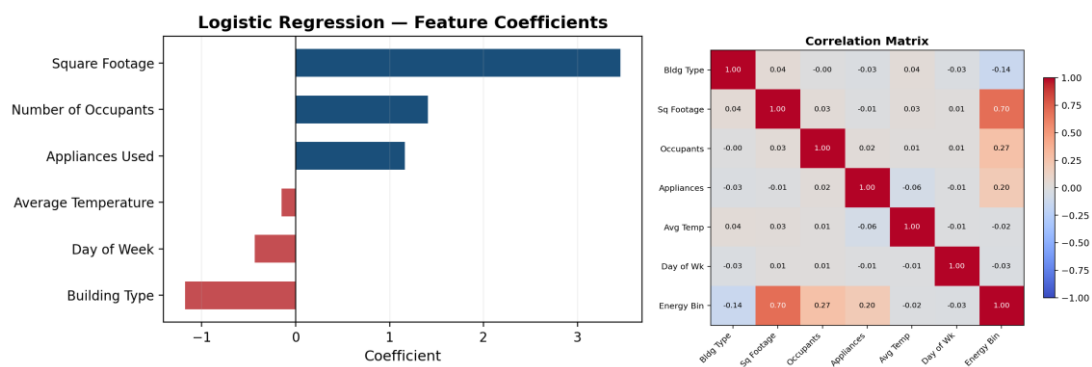


Figure 6. Correlation matrix of all features and the binary energy target.

Key findings: Square Footage dominates (coeff 3.46), far exceeding all others. Occupants (1.41) and Appliances (1.16) are strong secondary predictors. Building Type (-1.18) is negative—Residential consumes less. Avg Temperature (-0.15) is negligible, consistent with its non-significant p-value.

No multicollinearity: All VIF ≈ 1.0 . No confounders detected (no feature pair has $|r| > 0.3$ with both target and another feature).

7. Conclusion

1. Excellent classification. All classifiers achieve AUC > 0.95 and 91% accuracy. LR, LDA, and QDA achieve perfect precision (1.000).

2. LDA $\approx$ QDA. AUC 0.960 vs 0.965—the small gap confirms a linear boundary is largely sufficient, though QDA's flexibility provides marginal gain.

3. Optimal thresholds matter. Youden's J statistic yields thresholds of 0.764 (LDA) and 0.788 (QDA), outperforming the naive 0.5 default.

4. Multi-class reveals structure. Low (AUC=0.951) and High (AUC=0.916) are easy; Medium (AUC=0.758) overlaps both neighbors.

5. Linear beats Poisson. $R^2 = 0.814$ confirms energy is continuous, not count data.

6. Square Footage dominates. $Z = 14.76$, $p \approx 0$, odds ratio 41.7 $\times$. No multicollinearity or confounders.

References

- [1] James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). An Introduction to Statistical Learning (2nd ed.). Springer.
- [2] <https://www.kaggle.com/datasets/govindaramsriram/energy-consumption-dataset-linear-regression>